# THE LOCUS WHERE A MORPHISM IS AT WORST NODAL IS A MAXIMAL OPEN SUBSPACE STABLE UNDER BASE CHANGE

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*Editorial note.* Counted span is the author's own proof environment, source lines 13381-13403, immediately following the statement at 13371-13379.

*Editorial note.* Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is not counted beside it.

*Editorial note.* Two same-chapter lemmas and one cross-chapter lemma are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- The nodal locus in the case of schemes. Algebraic Curves, lemma curves-lemma-locus-where-nodal, cited at proof lines 13390-13391. Supplies the maximal open subscheme U' of the etale cover over which the morphism is at-worst-nodal of relative dimension 1.
- Local description of nodal morphisms. chapter lemma lemma-nodal-local, cited in the proof. Provides the local criterion used to show the descended open is at-worst-nodal.
- Etale descent of open subspaces. used at proof lines 13389-13393 (U -> X surjective etale). Descends the open subscheme U' of the cover to an open subspace X' of X.

*Source reference.* Original reference 0DSC: focused-seeds/geometry/S9-spaces-locus-where-nodal-spaces/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0DSG: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13320–13320. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$ which is flat and locally of finite presentation. Then there is a maximal open subspace $X' \subset X$ such that $f|_{X'} : X' \to Y$ is at-worst-nodal of relative dimension 1. Moreover, formation of $X'$ commutes with arbitrary base change.*

*Proof.* Choose a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

where $U$, $V$ are schemes, the vertical arrows are étale, and $U \to X$ is surjective. By the lemma for the case of schemes (Algebraic Curves, Lemma 0DSC) we find a maximal open subscheme $U' \subset U$ such that $h|_{U'} : U' \to V$ is at-worst-nodal of relative dimension 1 and such that formation of $U'$ commutes with base change. Let $X' \subset X$ be the open subspace whose points correspond to the open subset $\mathrm{Im}(|U'| \to |X|)$. By Lemma 0DSG we see that $X' \to Y$ is at-worst-nodal of relative dimension 1 and that $X'$ is the largest open subspace with this property (this also implies that $U'$ is the inverse image of $X'$ in $U$, but we do not need this). Since the same is true after base change the proof is complete. $\square$